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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CC2002-1 – INTERNET OF THINGS
(For CCE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Cloud services in IoT primarily add value by:
A) Centralizing computation, storage, and analytics B) Offering universal addressing for devices
C) Eliminating the need for gateways D) Restricting device heterogeneity
2. In IoT device design, energy efficiency is MOST critical for:
A) Cloud-based data centers B) Battery-operated remote sensors
C) Smart home hubs with constant power D) High-performance industrial robots
3. Which of the following describes an IoT "actuation loop"?
A) Data → Cloud → Human decision → Device command B) Gateway → Network → Storage → Analytics
C) Sensor data → Processing → Actuator response D) Actuator → Sensor → Gateway → Cloud
4. A smart home IoT system integrates lights, heating, and appliances. Which principle ensures it can later support security cameras without redesign?
A) Reliability B) Redundancy
C) Minimal latency D) Modularity
5. In Raspberry Pi, which interface is commonly used for high-speed data transfer?
A) GPIO B) I2C
C) SPI D) USB
6. Which communication protocol is suitable for IoT devices to communicate over long distances?
A) MQTT B) CoAP
C) ZigBee D) LoRaWAN
7. What type of I/O interface does an Arduino use to read analog signals?
A) Digital pins B) Analog pins
C) Serial ports D) PWM pins
8. What is the typical voltage level for digital I/O pins on an Arduino Uno?
A) 1.8V B) 3.3V
C) 5V D) 12V
9. What is the primary function of the CoAP protocol?
A) To provide secure communication B) To offer lightweight communication over UDP
C) To handle real-time data with high reliability D) To support high-bandwidth applications



10. Which of the following protocols is designed for low-power and low-bandwidth communication in IoT?
 - A) MQTT
 - B) TCP
 - C) CoAP
 - D) HTTP
11. What does ZigBee primarily use for communication between devices?
 - A) RF (Radio Frequency)
 - B) TCP
 - C) UDP
 - D) HTTP
12. What is the primary role of an MQTT broker in IoT device integration?
 - A) To store data
 - B) To manage communication
 - C) between devices
 - D) To analyze data
13. In a cloud-based IoT system, which service is used for data integration from multiple sources?
 - A) Data Warehousing
 - B) Data Lake
 - C) Data Integration Platform
 - D) Data Mining
14. What kind of data is best stored in a NoSQL database?
 - A) Structured data with complex relationships
 - B) Large volumes of unstructured or semi-structured data
 - C) Real-time transactional data
 - D) Data requiring complex queries
15. Which component is responsible for managing the communication between ZigBee devices?
 - A) ZigBee Coordinator
 - B) MQTT Broker
 - C) HTTP Server
 - D) CoAP Server
16. In an IoT system, which API method is used for retrieving data from a device using HTTP?
 - A) POST
 - B) PUT
 - C) GET
 - D) DELETE
17. Which function is used in Arduino to initialize serial communication?
 - A) begin()
 - B) setup()
 - C) start()
 - D) init()
18. To display sensor readings on an OLED screen using Arduino, which library is commonly used?
 - A) Adafruit_GFX
 - B) Liquid Crystal
 - C) TFT
 - D) Wire
19. How can an Arduino program detect a button press?
 - A) By using an external library
 - B) By checking for a low signal on the output pin
 - C) By measuring the resistance of the button
 - D) By checking for a high signal on the input pin
20. For controlling a motor with a push button using Arduino, what is required to handle the motor's power?
 - A) Capacitor
 - B) Relay
 - C) Diode
 - D) Transistor

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe three real-world IoT applications and their benefits.	5	L*2	1	1,2
	b) Explain the role of sensors and actuators in a smart home system	5	L2	1	1,2
	c) How does Machine-to-Machine (M2M) communication relate to IoT?	6	L2	1	1,2
2.	a) Why is data management a challenge in IoT systems?	5	L2	1	1,2
	b) How does IoT integration improve decision-making in businesses?	5	L2	1	1,2
	c) Discuss the role of the cloud in enabling IoT services and applications.	6	L2	1	1,2

CC2002-1**SEE – December 2025**

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|-------|---|---|----|---|-----|
| 3. a) | Write an Arduino program to interface an ultrasonic sensor. The program should measure the distance of an object and display it on the Serial Monitor. | 5 | L3 | 2 | 1,2 |
| b) | Write an Arduino program to interface a soil moisture sensor. The program should read the soil moisture level using an analog pin and display it on the Serial Monitor. | 5 | L3 | 2 | 1,2 |
| c) | Write an Arduino program to interface a smoke sensor (MQ-2). The program should read the sensor value and display a warning message if smoke is detected. | 6 | L3 | 2 | 1,2 |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | What is Communication protocol? Explain the key features in MQTT | 5 | L2 | 3 | 1,2 |
| b) | List and explain the main components of ZigBee. | 5 | L2 | 3 | 1,2 |
| c) | Mention the strengths of Bluetooth in IoT applications. | 6 | L2 | 3 | 1,2 |
| 5. a) | With a neat sketch, explain the traditional ZigBee stack. | 5 | L2 | 3 | 1,2 |
| b) | Explain the CoAP message format and its significance in IoT. | 5 | L2 | 3 | 1,2 |
| c) | Differentiate between Bluetooth, wifi and NFC. | 6 | L2 | 3 | 1,2 |
| 6. a) | Describe the working of the Push–Pull communication model in IoT. | 5 | L2 | 4 | 1,2 |
| b) | Explain the Exclusive Pair communication model with an example. | 5 | L2 | 4 | 1,2 |
| c) | List the key components of the IoT solution framework. | 6 | L2 | 4 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | Explain how IoT- enabled sensors improve quality control in manufacturing processes. | 8 | L2 | 5 | 1,2 |
| b) | Explain how IoT-based traffic monitoring systems enhance commuter safety. | 8 | L2 | 5 | 1,2 |
| 8. a) | Apply IoT solutions to suggest how smart irrigation can reduce water wastage in drought-prone areas. | 8 | L2 | 5 | 1,2 |
| b) | Define remote patient monitoring and give one IoT device used in it. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
